WEEK 9-POWER BI TASK

REAL ESTATE DATA CLEANING AND TRANSFORMATION USING POWER BI

1.Introduction

The Week 9 project focuses on cleaning and transforming real estate data using Power BI. The King County House Sales dataset was used to perform data preparation activities in Power Query Editor. The main purpose was to organise the dataset and prepare it for further analysis and dashboard development.

2. Objectives

1. To import the real estate dataset into Power BI.

2. To clean and filter unnecessary records.

3. To extract Year and Month from the Date column.

4. To create custom columns for data analysis.

5. To classify properties based on their living area.

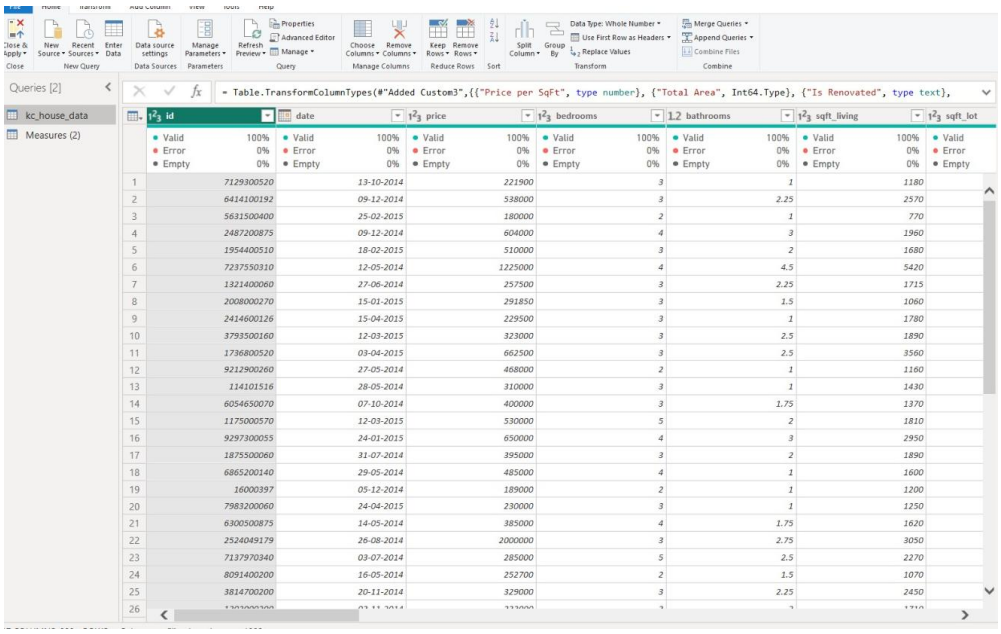
6. To verify and modify data types.

3. Tools and Technologies Used

- Power BI Desktop
- Power Query Editor
- King County House Sales Dataset

4. Data Import

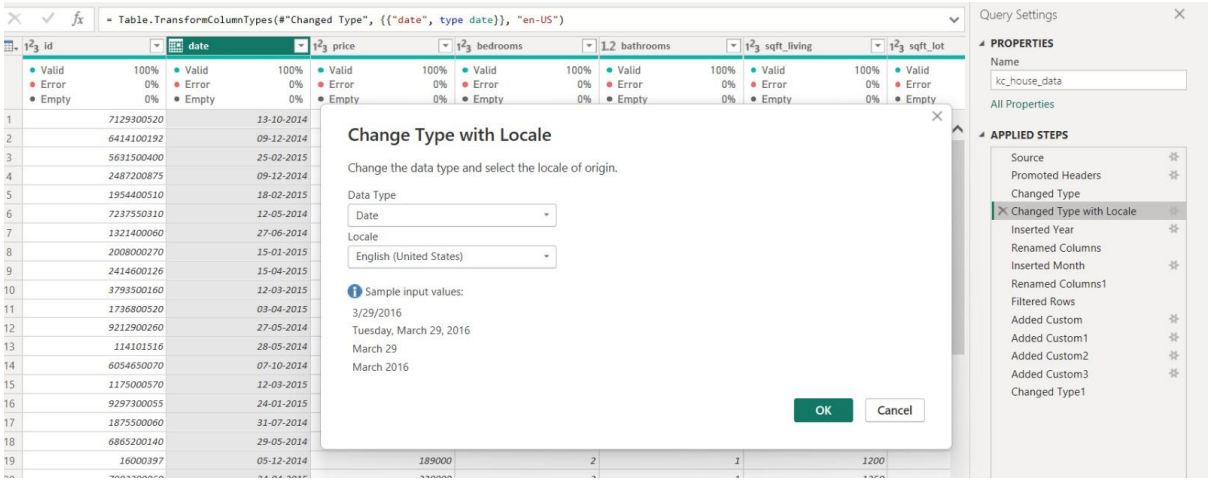
The King County House Sales dataset was imported into Power BI. The dataset was opened in Power Query Editor to perform the required data cleaning and transformation operations.



5. Data Cleaning and Transformation

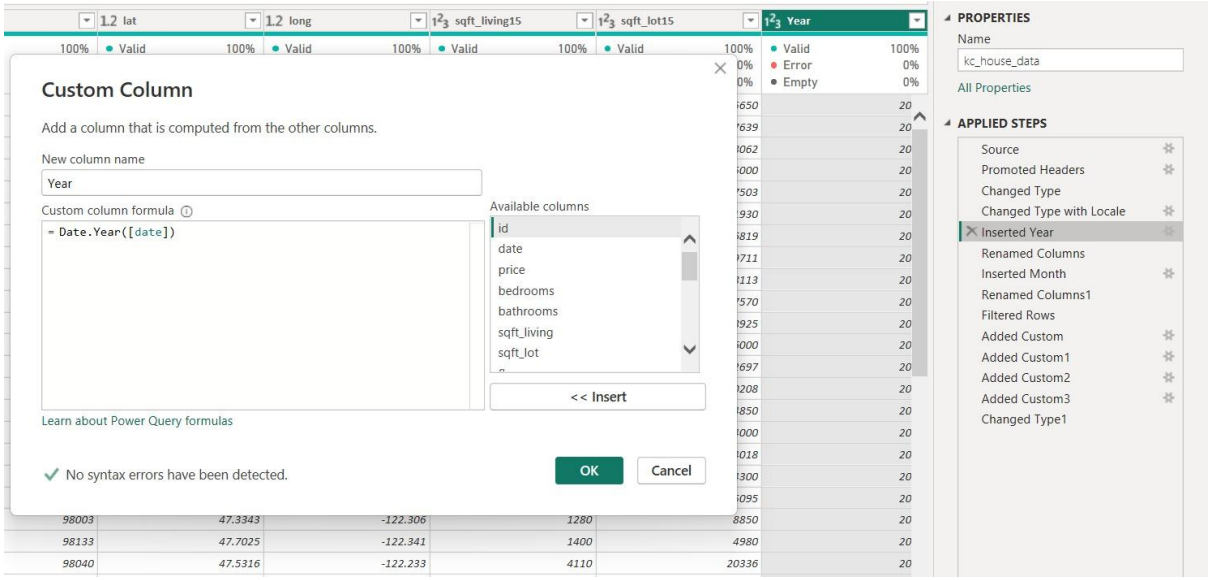
Step 1: Change Data Types

The data types of the columns were checked and modified according to their values. Locale settings were applied where necessary for correct date conversion.



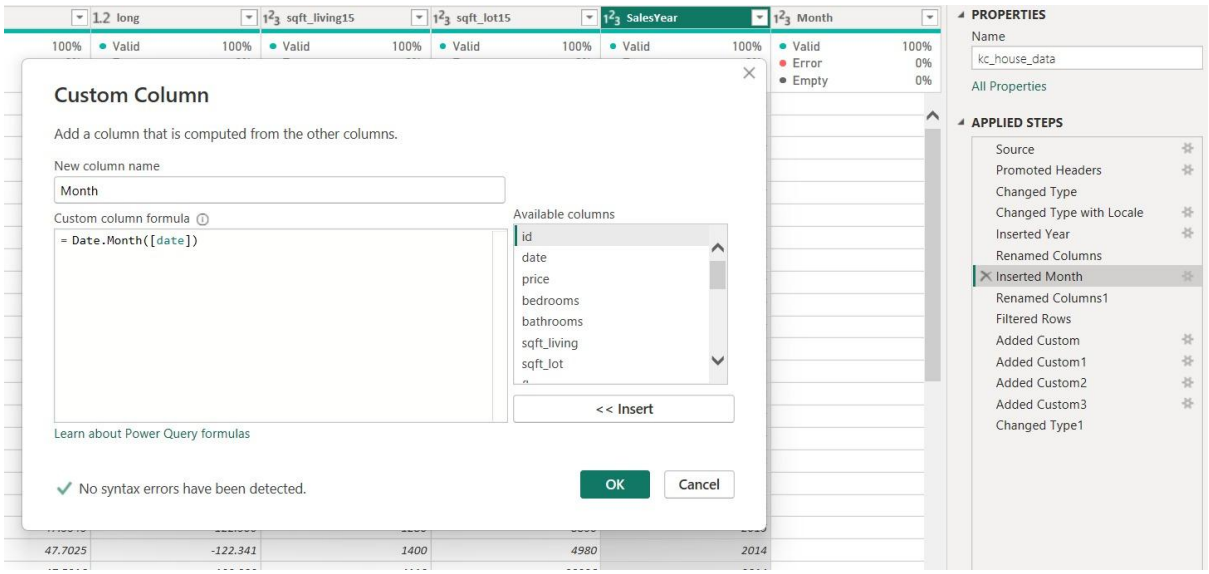
Step 2: Extract Year

The Year value was extracted from the Date column using the Date transformation option. The newly created column was renamed as Year.



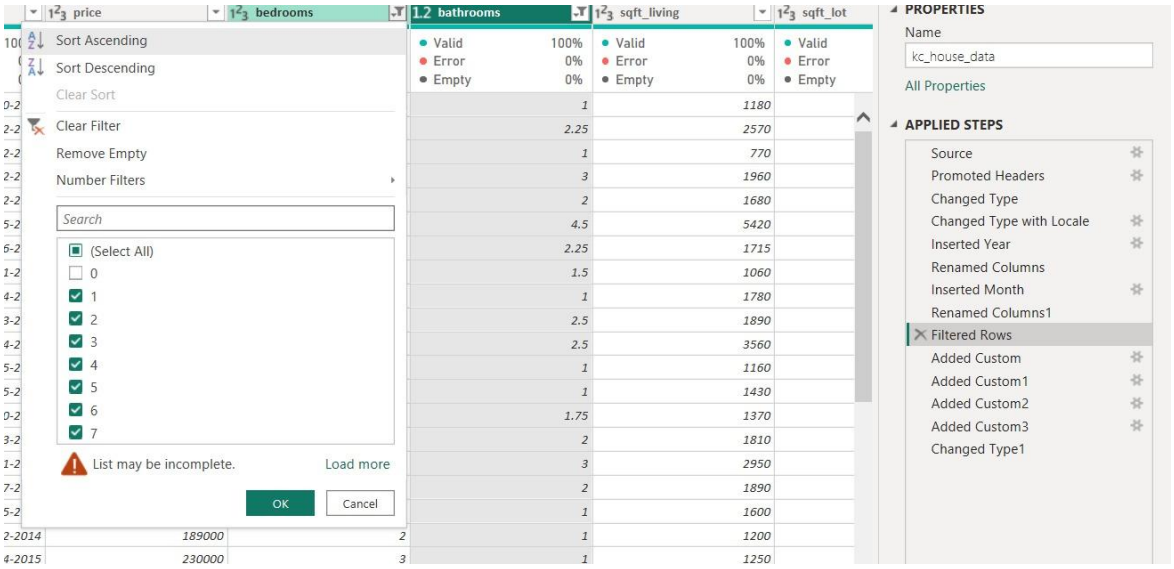
Step 3: Extract Month

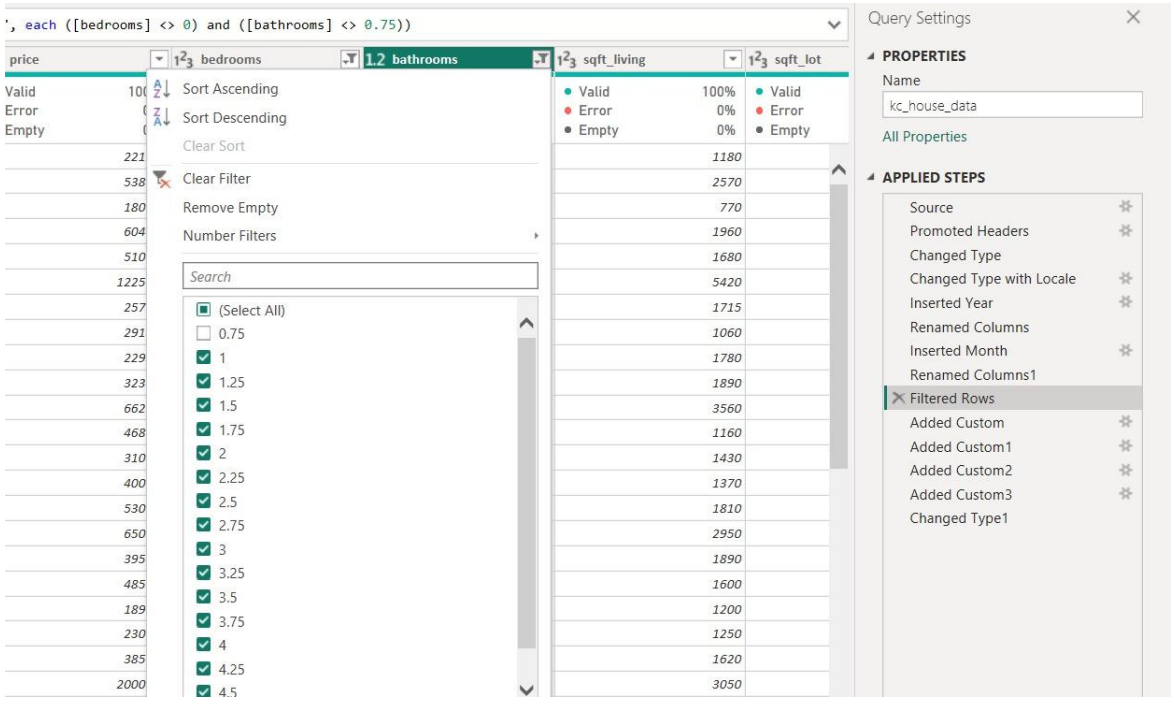
The Month value was extracted from the Date column. The newly created column was renamed as Month.



Step 4: Filter Rows

Unnecessary records containing zero bedrooms and zero bathrooms were filtered out to improve the quality of the dataset.





Filtering Unnecessary Records

Step 5: Create Price per SqFt Column

A custom column named Price per SqFt was created to calculate the property price per square foot of living area.

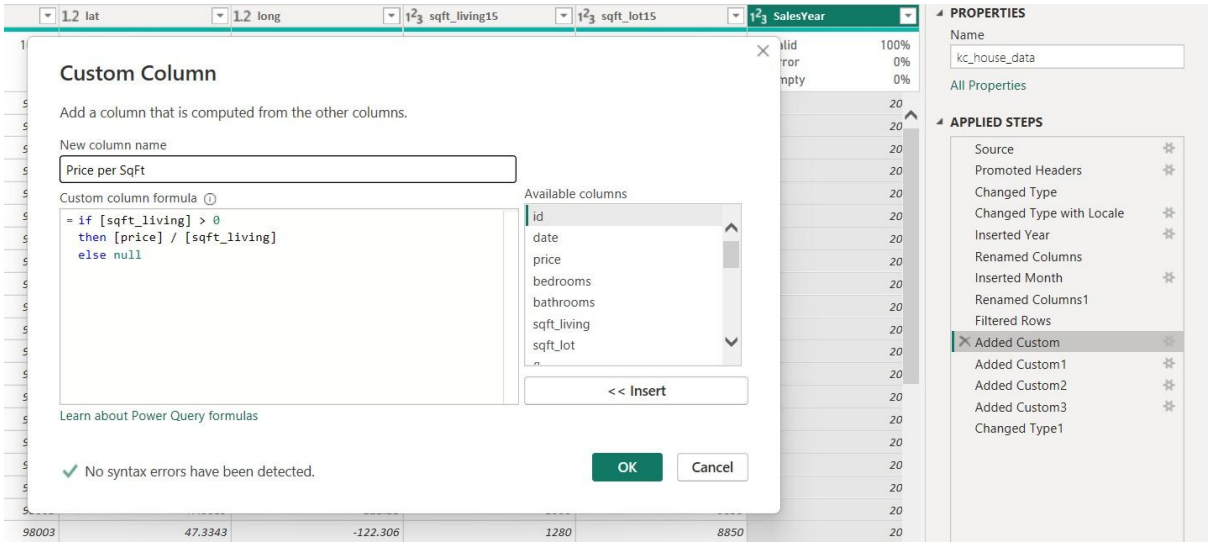
Formula:

Price per SqFt=

if [sqft_living] > 0

then [price] / [sqft_living]

else null



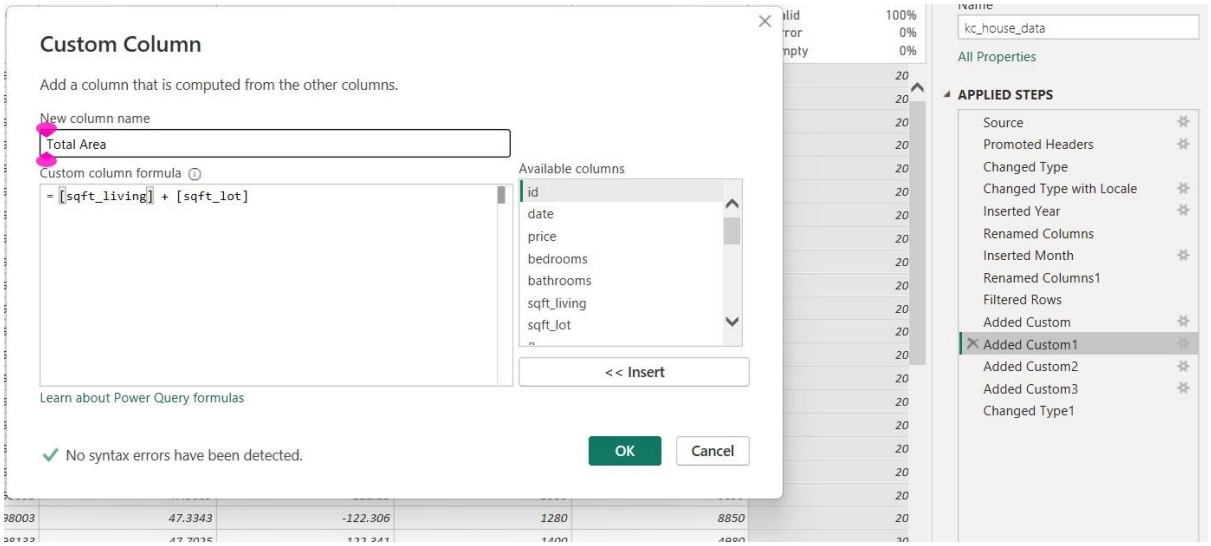
Step 6: Create Total Area Column

A custom column named Total Area was created by adding the living area and lot area of each property.

Formula

Total Area=

[sqft_living] + [sqft_lot]



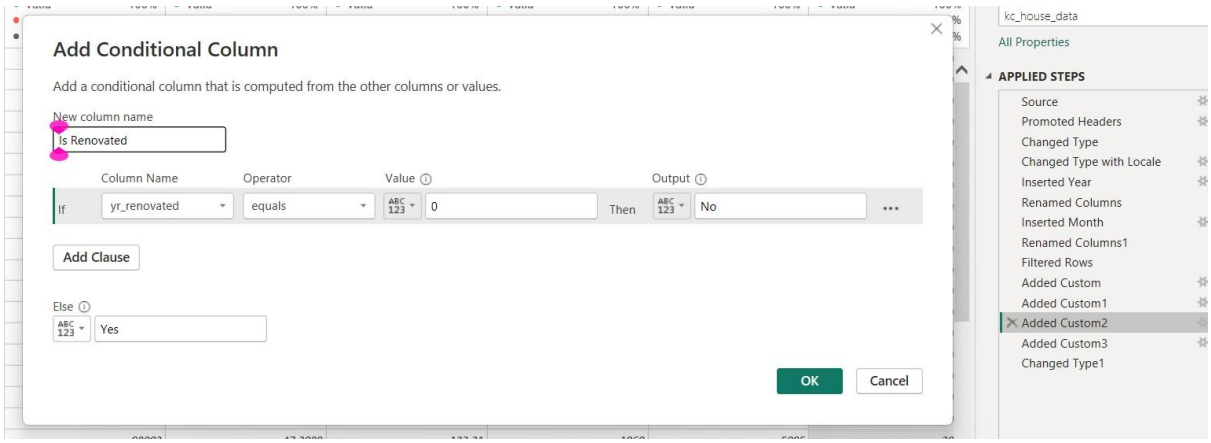
Custom Column Creation

Step 7: Create Is Renovated Column

A conditional column named Is Renovated was created using the yr_renovated column to identify renovated properties.

- If `yr_renovated = 0`, the result is No.

- Otherwise, the result is Yes.



Renovation Status Classification

Step 8: Create House Size Column

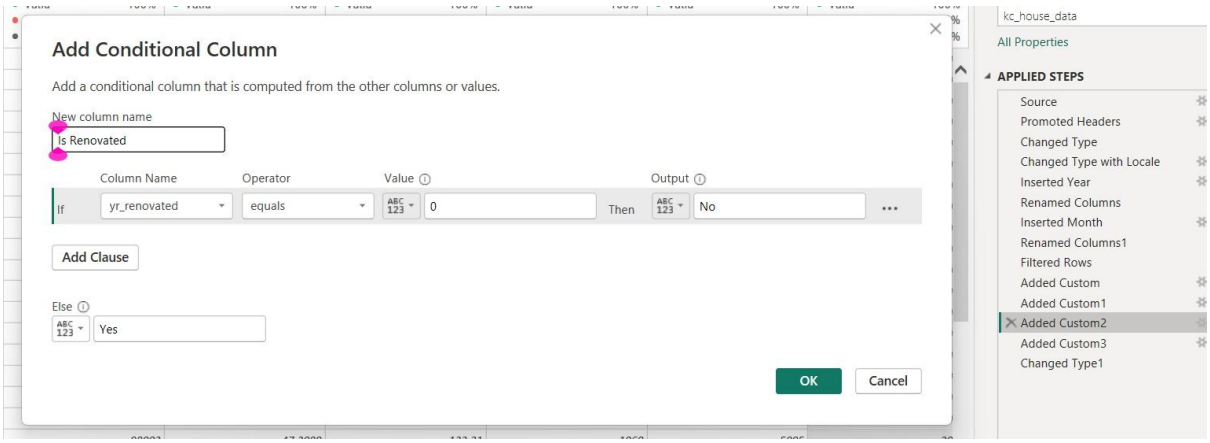
A conditional column named House Size was created to classify properties based on their living area.

- Below 1000 sq. ft. – Small

- Below 2000 sq. ft. – Medium

- Below 4000 sq. ft. – Large

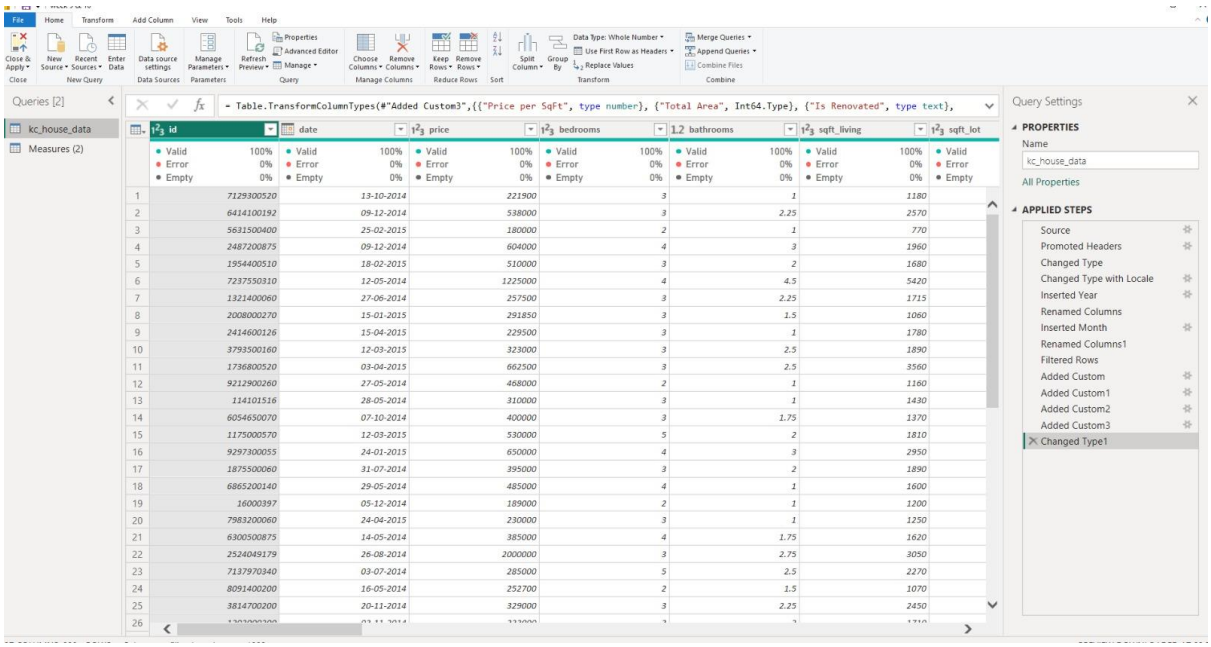
- 4000 sq. ft. and above – Luxury



House Size Categorisation

Step 9: Final Data Type Modification

The data types of the newly created columns were checked and corrected where necessary. Numeric columns were assigned suitable numeric data types, while categorical columns were set as text.



Final Transformed Dataset

6. Final Dataset Preparation

After completing all the cleaning and transformation activities, the dataset was reviewed to ensure that the columns contained appropriate values and data types. The newly created columns were verified, and the transformed dataset was prepared for further analysis and dashboard development in Week 10.

7. Conclusion

The Week 9 Real Estate Data Cleaning and Transformation project was successfully completed using Power BI. The dataset was cleaned, organised, and transformed using Power Query Editor. The prepared dataset provides a foundation for developing an interactive Real Estate Analytics Dashboard in Week 10.